

Heterogeneous Defect Prediction using Federated Prototype Learning

Implementation and Experimental Analysis

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Problem Statement

- Software defect prediction identifies faulty modules early
- Cross-project prediction is difficult because:
 - Different projects use different metrics
 - Defect data is highly imbalanced
- Sharing Data across organizations is often not allowed
- **Problem:** How to perform cross-project defect prediction without sharing raw data?

- Single-project data is often:
 - Small
 - Imbalanced
 - Noisy
- Cross-project learning improves prediction quality
- Privacy constraints restrict data sharing
- Need a method that:
 - Preserves privacy
 - Works across heterogeneous projects

Existing Work and Research Gap

- Traditional defect prediction:
 - Requires centralized data
 - Performs poorly under heterogeneity
 - Not privacy preserving
- Federated Learning approaches:
 - Shares gradients or model parameters
 - Sensitive to noisy and imbalanced data
- **Research Gap:** Lack of robust federated learning methods for heterogeneous defect prediction

Challenges

- Metric heterogeneity: Different feature sets across different projects
- Class imbalance: Very few defective modules
- Label noise: Inconsistent defect labels
- Federated learning challenge:
 - Gradient-based aggregation performs poorly under heterogeneity

- Adopted Federated Prototype Learning (FPLPA)
- Key idea:
 - Share class prototypes instead of model parameters
- Workflow:
 - Data preprocessing and noise filtering
 - Feature selection using Chi-square test
 - Local CNN model training
 - Local class prototype computation
 - Prototype aggregation on the server
 - Prototype based regularization

Experimental Setup

- Datasets Used:
 - NASA: PC1, PC3, PC4, MW1, CM1
 - ReLink: Apache, Safe, ZXing
- Each dataset acts as one federated client
- Evaluation Metrics:
 - AUC
 - G-Mean
- Aggregation Strategy Tested:
 - Full client aggregation
 - Random subset aggregation-3,5,8 clients

Results and Observations

- Results (Full Aggregation):
 - Average AUC: 0.59
 - Average G-Mean: 0.39
- Observations:
 - Stable training behavior
 - Performance affected by:
 - Noisy clients
 - Severe class imbalance
 - Indicates full aggregation is not always optimal

- Introduce weighted prototype aggregation
- Assign weight to clients based on - Data quality, Noise level, Local performance
- Design a server side selection algorithm
 - Prefer reliable clients during aggregation
- Improve robustness against noisy and imbalanced datasets
- Extend the evaluation to more datasets

- Heterogeneous Defect Prediction Based on Federated Prototype Learning **IEEE Paper**
- NASA Software Defect **Datasets**
- AEEEM Software Defect **Datasets**
- ReLink Software Defect **Datasets**
- Federated Learning **Literature**

Thank You!